

Assignment AI

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1. Stages in data analytics lifecycle are:

(i) Data Acquisition: Gathering data from various sources ensuring it's relevant to objectives of analysis.

(ii) Data Preparation: Cleaning, organizing and transforming the data to make it ready for analysis. This includes handling missing data or values, managing outliers and standardizing formats.

(iii) Exploration: Conducting exploratory data analysis to uncover patterns, relationships and insights within the data. Tools and visualisation are often employed at this stage.

(iv) Analysis: Applying statistical, mathematical or machine learning technique to extract meaningful insights and build predictive or descriptive models.

(v) Visualization: Presenting results through graphs, charts and dashboards to communicate findings effectively.

(vi) Decision Making: Using insights and visualized outcomes to inform strategies, optimize processes or guide actions.

2. Handling missing values and outliers is a critical part of data cleaning stage.

Missing values:

- (i) Deletion: If missing data is minimal and doesn't significantly impact dataset, you can remove rows or columns with missing values.
- (ii) Imputation: Replacing missing values with statistical measures like mean, median or mode. For categorical data we use most frequent category.
- (iii) Prediction: Use ML models to predict missing values which can be useful for certain predictions based on other features in dataset.
- (iv) Flagging: Create new feature to indicate missing values which can be useful for certain analyses.

Outliers:

- (i) Removal: If outliers are errors or irrelevant, you can exclude them from dataset.
- (ii) Transformation: Apply mathematical transformation (log or sqrt) to reduce impact of outliers.
- (iii) Capping: Replace outliers with maximum or minimum threshold value.
- (iv) Robust models: Use algorithms that are less sensitive to outliers such as decision trees or ensemble methods.

3. Exploratory Data Analysis is a crucial process of examining and visualizing data to summarize its main characteristics often with help of visual methods. It is a crucial step in data analytics process that helps analysts understand what data shows before applying any model or formal statistical methods.

Key activities in EDA:

- i) Understanding data structure
- ii) Summary Statistics
- iii) Data Visualization
- iv) Correlation analysis
- v) Outlier Detection

Importance of EDA:

- i) Improves data quality - Detects and corrects data errors, missing values and anomalies
- ii) Guides model selection - Helps on selecting right models based on data patterns
- iii) Uncovers patterns and trends - Reveals hidden insights and variable relationship.
- iv) Reduce risks - Avoid incorrect assumption and ensures validity of analysis.
- v) Better Communication - Visuals make it easier to explain insights to stakeholders.

④ Types of Data Analytics	What it does	Key Questions Answered	Examples
Descriptive analysis	Summarize past data to understand what happened	What happened?	A monthly sales report showing total revenue and no. of products sold.
Diagnostic analysis	Examines data to find out why something happened	Why did it happen?	Analysing sales dip in Feb by looking into marketing spend, customer feedback and campaign activity.
Predictive analysis	Use statistical model and ml to forecast future outcomes	What is likely to happen?	Predicting customer churn based on usage patterns and support interactions.
Prescriptive analysis	Recommends action to take for optimal outcomes	What should we do?	Suggesting personal promotion to certain high value customers and reduce churn.

⑤ Data visualization aids in interpretation stage of data analytics in following ways:

- (i) Simplifies complex data
- (ii) Reveals patterns and trends
- (iii) Supports better decision-making
- (iv) Improves communication
- (v) Highlights key insights quickly
- (vi) Validates analytical results
- (vii) Engages audience.
- (viii) Enhances comparisons.

Example: Instead of showing table of monthly sales, a line graph instantly shows whether sales are rising or falling, making trends clear at a glance.

6 Data Type	Classification	Explanation
CSV files	Structured	Data in CSV files is organized in rows and columns, making it easily searchable.
Social media posts	Unstructured	Posts often include text, images and videos without predefined format.
Json data	Semi-Structured	Json has a flexible structure with tags and meta data, but lacks a rigid schema.
Relational database tables	Structured	Data is stored in tabular format with predefined schema.
Email Content	Unstructured	Emails contain diverse elements like text, attachments and metadata.

⑦

Propositional logic:

- Propositional Calculus or boolean logic
- deals with propositions, which are statements that can either be true or false

→ Basic components are:

- Propositions → statement which are either true or false
- Logical connectives → Operators such as And ($\wedge$), OR ($\vee$), Not ($\neg$), implies ($\rightarrow$), biconditional ($\leftrightarrow$) used to combine propositions.

iii) Truth values: Each proposition has truth value of either true (T) or false (F)

For example

P: "It is raining"

Q: "The ground is wet"

In propositional logic it is

$(P \rightarrow Q)$, if it is raining, then ground is wet.

⑧ First Order Predicate logic or first order logic

- Extends propositional logic by introducing quantifiers and predicates
- deals with objects, properties and relationships.
- more expressive representation.

Basic components:

① Constants: specific objects in domain (eg. Alice, Bob)

② Variables: Symbol which represent any object in domain. (eg. x, y)

③ Predicates: functions that map objects to truth values.
eg: likes (Alice, Icecream)

④ Quantifiers: Symbols indicate scope of a statement eg: $\forall$ (forall), $\exists$ (exists)

⑤ Logical connectives: Same as in propositional logic

for eg: likes (x, y): "x like y"

$\forall x \exists y (\text{likes}(x, y))$ [For every person x, there exist a person y such that x likes y]

⑥ Heuristic is a function that estimates cost to reach the goal from current node.
→ denoted by $h(n)$ where n is current node.

A*

$$f(n) = g(n) + h(n)$$

$g(n)$ = actual cost from

Start node to node n

$h(n)$ = estimate cost from node n to goal node

A* considers both the actual cost so far and estimated cost to go.

Is efficient and slower because it is more careful but it is accurate.

A* provides complete solution.

GBFS.

$$f(n) = h(n)$$

where $h(n)$ is estimated cost from node n to goal node

Only considers estimate cost to goal, ignoring the cost to goal, ignoring cost already spent.

GBFS is faster but it is riskier.

GBFS is incomplete because it can get stuck in infinite loop

→ A* guarantees optimality with admissible heuristic.

→ A heuristic $h(n)$ is admissible if:

$h(n) \leq h^*(n)$ where $h^*(n)$ is actual cost to reach the goal from n .

→ A* combines $g(n)$ and $h(n)$, ensuring it explores all cheaper path before more expensive ones.

→ $h(n)$ doesn't overestimate, A* never misses cheaper path, just guaranteeing optimality.

→ GBFS only uses $h(n)$ If heuristic is misleading (eg makes a long path look shorter) GBFS may follow a suboptimal path. It doesn't amount for how much cost has already been incurred ($g(n)$).

Key Differences

Feature	Propositional Logic	First Order Logic
Basic Unit	Propositions	Predicates, constant, variables
Expressiveness	Limited to boolean values only	Expressive in representing relationships and properties
Quantifiers	No quantifiers	Universal ($\forall$) & existential ($\exists$)
Syntax	Combines propositions using logical connectives	Use predicates and quantifiers
Semantics	Truth tables	Interpretation over a domain
Use Cases	Simple problem (eg: Circuit design, rule based system)	Complex problems (eg: AI, reasoning, ontology modeling)
Example	$P \rightarrow Q$	$\forall x \exists y (\text{Likes}(x, y))$

Example where FOL is more expressive than propositional logic.

S₁: Student 1 submitted the assignment

S₂: " 2 " " "

S₃: " 3 " " "

⋮
⇒ $S_1 \wedge S_2 \wedge \dots \wedge S_n$

if FOL:

⇒ $\forall x (\text{Student}(x) \rightarrow \text{Submitted}(x, \text{Assignment}))$

⇒ for all x if x is a student, then x has submitted the assignment.

→ This is compact and general for all.

Q9 Represent FOL

(i) "All humans are mortal."

$$\forall x (\text{Human}(x) \rightarrow \text{mortal}(x))$$

for all x , if x is human then x is mortal.

(ii) "Socrates is a human."
 $\text{Human}(\text{Socrates})$

(iii) "Therefore, Socrates is mortal."
 $\text{mortal}(\text{Socrates})$.

from (ii) and (i):

$$\text{Human}(\text{Socrates}) \therefore \text{mortal}(\text{Socrates})$$

Q10 Hypothesis testing is a method (statistical method) used to make decision or influence about a population based on a sample of data.

→ A hypothesis is a claim or assumption about a population.

→ Hypothesis testing uses sample data to decide which one is more likely to be correct.

Types of hypothesis

① Null hypothesis (H_0):

→ Starting assumption in statistics

→ It says there is no relationship between groups

→ It is the claim that nothing has changed or there is no effect.

eg - A new drug doesn't work → H_0 : The drug has no effect
or A win in favor → H_0 : $P(\text{Heads}) = 0.5$

② Alternative hypothesis (H_1 or H_a):

- This is what you hope to prove - that there is a difference or an effect or a change
- If evidence is strong enough, you reject H_0 in favour of H_1
ex - coin is biased → $H_1: P(\text{heads}) \neq 0.5$
The drug works → H_1 : drug has an effect.

Steps in hypothesis testing

① State hypothesis (H_0 and H_1)

② Choose significance level (α).

→ Common choice: 0.05 or 0.01

→ This is probability of rejecting H_0 when it actually true (Type I error)

③ Collect data and calculate test statistic like (z-test, chi square etc.)

④ Calculate p-value

↳ the probability of observing data if H_0 is true

⑤ Compare p value with α .

if $p \leq \alpha$ reject H_0 → significant result

if $p > \alpha$ fail to reject H_0 → not enough evidence

T-test: A statistical test used to compare averages of two groups to see if they are significantly different from each other.

g. gp1: 30 student who studied with method A.

gp2: 30 student who studied with method B.

here T-test is used to check if there is significant difference in avg test scores b/w two.

The hypothesis test here

Does people who drink coffee actually work faster?

H_0 : No effect of coffee both groups have same working time. $\mu_1 = \mu_2$

H_1 : Coffee drink people work faster $\mu_1 < \mu_2$

Here both group are independent

→ Independent Samples T-Test will work here

→ If on testing $p\text{-value} < 0.05$;

reject H_0 → Coffee really increases productivity
else Vice-versa

11. Purpose: Association rule mining is a data technique used to discover interesting relationship patterns or correlations among items in large datasets. Its primary purpose is to find rules that predict occurrence of an item based on occurrences of other items in same transaction or dataset.

Application in Market basket analysis → MBA is one of most common application of Association rule mining.

Goal: To understand customer purchasing behaviour by identifying products frequently bought together.

e.g. {Bread, Butter} → {Jam}

This means person who buy bread butter, also buy jam.

Key metrics in Association rule:

Support → Frequency of itemset in dataset (B)

Confidence → Likelihood of purchasing an item when given items are bought. ($A \rightarrow B$)

Lift → How much more likely A is bought with B than alone.
Confidence ($A \rightarrow B$) / Support (B) If lift > 1 the association.

Real world application -

- Product placement
- Cross selling
- Personalized Recommendations.

b. mean of temp = $\frac{20+22+25+24+26+23+21}{7}$
 $\approx \frac{161}{7} = 23^\circ\text{C}$

13. find mode.

2, 3, 4, 2, 2, 5, 3, 2, 4, 3, 3

freq. $2 \rightarrow 3$ times
 $3 \rightarrow 4$ "
 $4 \rightarrow 2$ "
 $5 \rightarrow 1$ "

mode $\hat{=}$ highest freq. = 4 times = 3 shown

14. fundamental differences:

Supervised Learning

- $\rightarrow$ learns from labelled data.
- $\rightarrow$ Predict output (label) based on input data
- $\rightarrow$ Output/Target labels are known
- $\rightarrow$ It's predictive

example algorithm are

- $\rightarrow$ Linear Regression
- $\rightarrow$ Decision trees
- $\rightarrow$ Support vector machines
- $\rightarrow$ Neural networks

Real world application

- $\rightarrow$ Spam detection
- $\rightarrow$ Medical diagnosis
- $\rightarrow$ Stock price prediction
- $\rightarrow$ Credit Scoring

Unsupervised Learning.

- $\rightarrow$ learns from unlabelled data.
- $\rightarrow$ Find hidden pattern or structures in data
- $\rightarrow$ No predefined labels.
- $\rightarrow$ It's descriptive.

example algorithm are

- $\rightarrow$ Kmeans Clustering
- $\rightarrow$ Hierarchical clustering
- $\rightarrow$ PCA (Principal component analysis)
- $\rightarrow$ Apriori (for association rules)

Real world application.

- $\rightarrow$ Customer segmentation
- $\rightarrow$ Anomaly detection
- $\rightarrow$ Market basket analysis
- $\rightarrow$ Recommender systems

- I. Understand dataset →
 - Review what kind of data is available.
 - Identifying columns like:
Age, gender, income, location (demographic), purchase,
amount, date, category, etc.
 - Understand what each column means

- II. data cleaning →
 - Check for missing values
 - inconsistent format checks
 - Check for outlier and duplicate entry
 - ensure data is clean and ready for analysis

- III. Descriptive Stats
 - Calculate basic statistics.
 - ↳ avg age
 - ↳ mean
 - ↳ mode

→ get a high level understanding of dataset.

- IV. Analyse Customer demographics.
How customer base is distributed by age, gender, income, regions

- V. Analysis purchasing behaviour →
 - How often do customers purchase
 - Avg purchase value
 - Any seasonal or monthly trends.

- VI. Identify patterns and relationships →
 - does income affects purchasing behaviour
 - is there gender based preferences?
 - Un cover meaningful insights for business decisions

- VII. Segment Customers →
 - Group customers based on behaviours
 - ↳ higher spenders vs lower spenders
 - ↳ frequent vs infrequent
 - ↳ new vs old.

VIII. Summarize key finding:
at end we should be able to answer

- who are our typical customers
- what do they buy
- when and how much do they buy

Possible questions can be:

- who are top most customers
- which product generates more revenue.
- which product is in loss?
- which region contributes more to sales etc.